

Exp No. 10: 5G Network Slicing Environment Using MATLAB

AIM: To analyze the performance of a 5G Network Slicing environment by studying the number of network slices, slice utilization, and end-to-end latency using MATLAB.

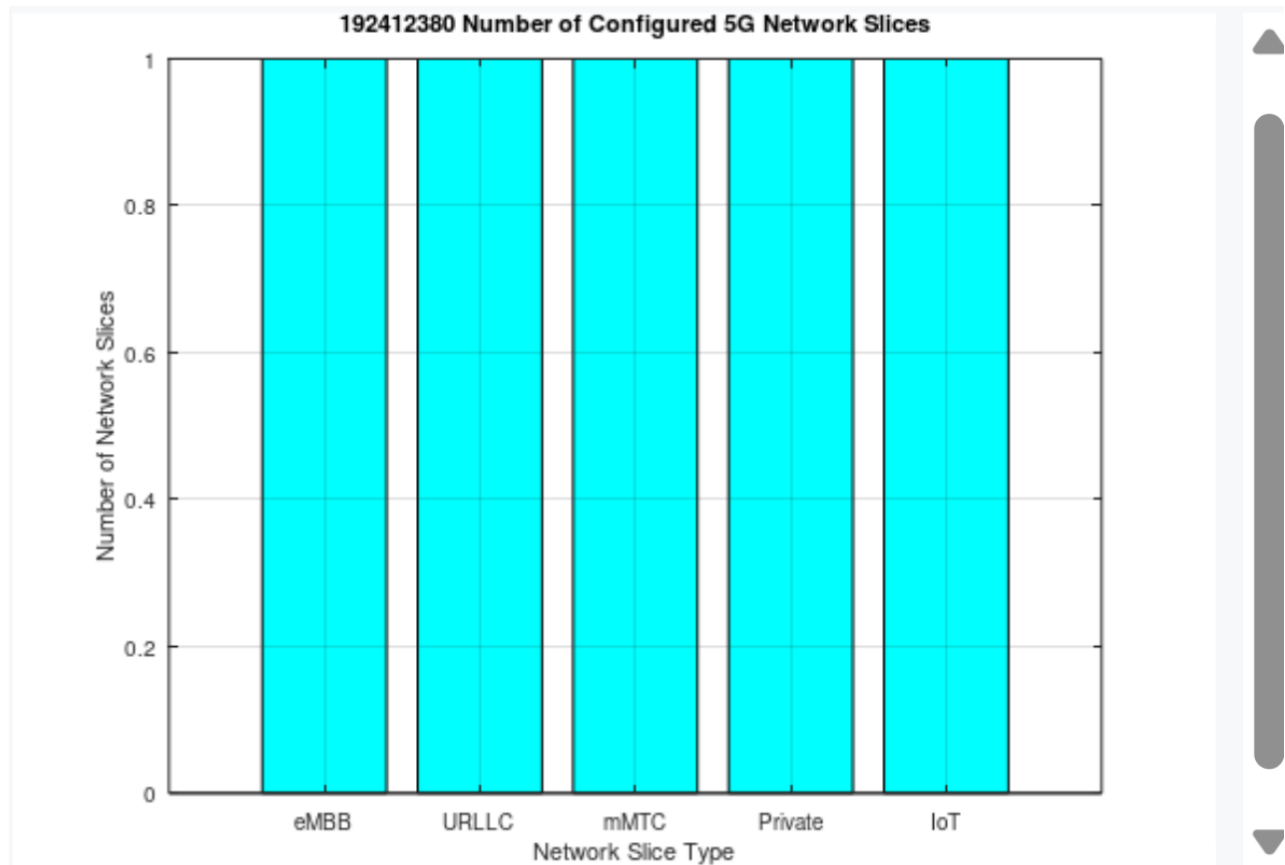
Objective 1:

Analyze the creation and allocation of multiple network slices in a 5G network using MATLAB. Observe how different slices can be configured to support diverse service requirements and applications.

Program :

```
1  clc;
2  clear;
3  close all;
4
5  % Network Slice Types
6  slices = {'eMBB', 'URLLC', 'mMTC', 'Private', 'IoT'};
7
8  % Number of Network Slices
9  count = [1 1 1 1 1];
10
11 figure;
12
13 % Bar Graph (Cyan Color)
14 bar(count, 'c');
15
16 grid on;
17
18 set(gca, 'XTick', 1:5);
19 set(gca, 'XTickLabel', slices);
20
21 xlabel('Network Slice Type');
22 ylabel('Number of Network Slices');
23 title('192412380 Number of Configured 5G Network Slices');
24
25 drawnow;
26 pause(5);
```

OUTPUT :



Objective 2:

Compare the utilization of different network slices using MATLAB. Analyze how efficiently network resources are allocated and utilized across multiple slices.

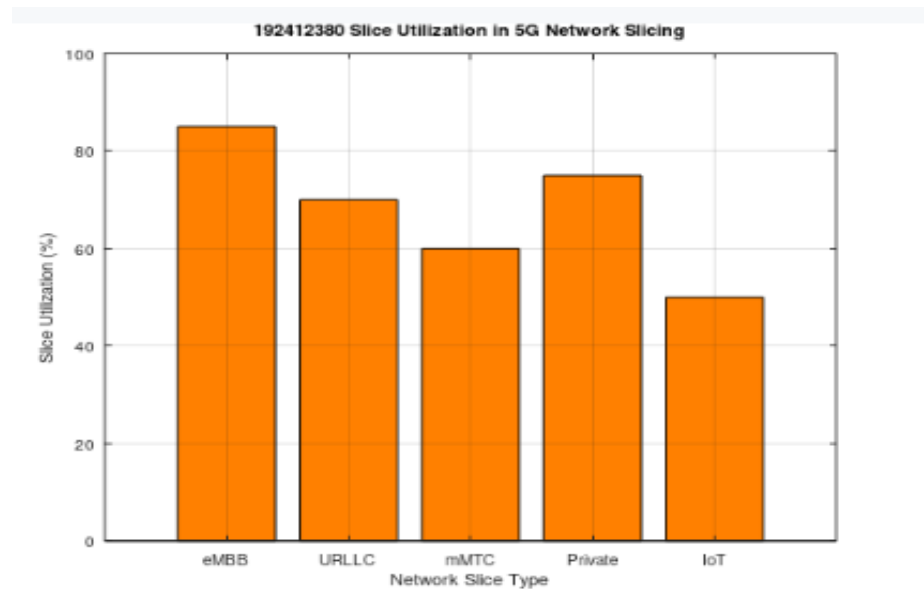
Program

```

1  clc;
2  clear;
3  close all;
4
5  % Network Slice Types
6  slices = {'eMBB','URLLC','mMTC','Private','IoT'};
7
8  % Slice Utilization (%)
9  utilization = [85 70 60 75 50];
10
11 figure;
12
13 % Bar Graph (Orange Color)
14 bar(utilization, 'FaceColor', [1 0.5 0]);
15
16 grid on;
17
18 set(gca, 'XTick', 1:5);
19 set(gca, 'XTickLabel', slices);
20
21 xlabel('Network Slice Type');
22 ylabel('Slice Utilization (%)');
23 title('192412380 Slice Utilization in 5G Network Slicing');
24
25 ylim([0 100]);
26
27 drawnow;
28 pause(5);

```

OUTPUT :



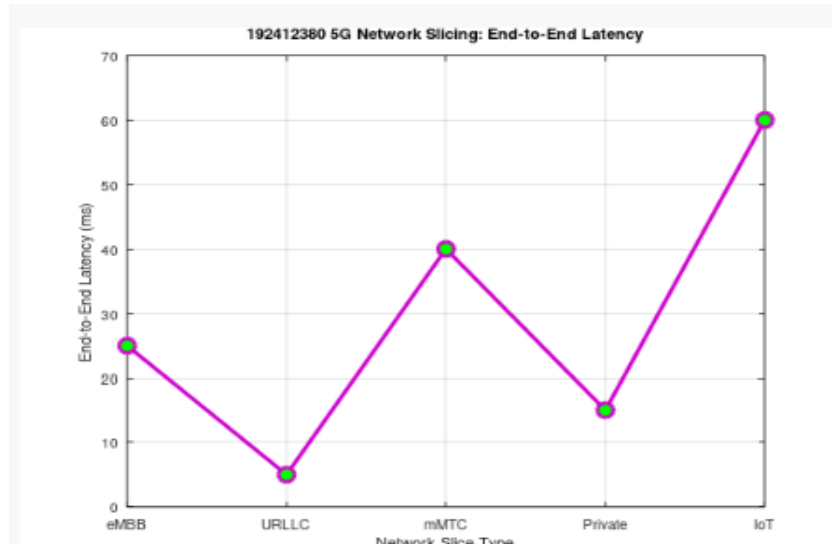
Objective 3:

Analyze the end-to-end latency of different network slices using MATLAB. Compare the latency values to understand the Quality of Service (QoS) requirements of various 5G applications.

program

```
1  clc;
2  clear;
3  close all;
4
5  % Network Slice Index
6  slices = [1 2 3 4 5];
7
8  % End-to-End Latency (ms)
9  latency = [25 5 40 15 60];
10
11 figure;
12
13 % Plot with Purple Line and Green Markers
14 plot(slices, latency, '-o', ...
15      'Color', [0.8 0 0.8], ...
16      'MarkerFaceColor', [0 1 0], ...
17      'LineWidth', 2, ...
18      'MarkerSize', 8);
19
20 grid on;
21
22 % Set X-axis labels
23 set(gca, 'XTick', 1:5);
24 set(gca, 'XTickLabel', {'eMBB', 'URLLC', 'mMTC', 'Private', 'IoT'});
25
26 xlabel('Network Slice Type');
27 ylabel('End-to-End Latency (ms)');
28 title('192412380 5G Network Slicing: End-to-End Latency');
29
30 ylim([0 70]);
31
32 drawnow;
33 pause(5);
34
```

OUTPUT :



Result:

The MATLAB analysis successfully demonstrated the characteristics of a 5G Network Slicing environment. The results showed that multiple network slices can coexist while exhibiting different utilization levels and latency characteristics based on their intended services.

Practical Questions

A 5G wireless communication system employs network slicing to partition a single physical network into multiple virtual networks, each optimized for different service requirements such as enhanced Mobile Broadband (eMBB), Ultra-Reliable Low-Latency Communication (URLLC), and massive Machine-Type Communication (mMTC). This enables efficient resource utilization and improved Quality of Service (QoS). Using MATLAB, simulate a 5G network slicing environment and analyze its performance.

1. To analyze the number of network slices created in a 5G network slicing environment and understand how different services are logically separated.
2. To compare the slice utilization (%) across different network slices and evaluate the efficiency of resource allocation in a 5G network.
3. To analyze the end-to-end latency (ms) of different network slices and study how latency varies based on service requirements in a 5G environment.